Dynamic Pricing for European Camping Group

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1 Problem and business context

European Camping Group (ECG) operates over 400 campsites across 10 European countries, each offering several accommodation types and serving four distinct market segments. With a 30-week open season, this implies more than 250,000 individual prices to set every year, one for every combination of campsite, accommodation, market, and arrival-week. We refer to such a combination as a *ROMGID* (reservable-option-market-group), the unit of analysis throughout the report. This project examines: can historical booking data be used to recommend better prices for these ROMGID's, and what would that be worth in revenue?

2 The trap

The most direct approach, regressing booked nights on price while controlling for accommodation and seasonality, yields an answer that is economically impossible. On a 50,000 row training sample, the estimated price coefficient is positive: $\beta = +0.29$, with the 80% credible interval $[+0.18, +0.42]$. Read literally, this means "raising prices increases bookings".

The reason is structural, not statistical. ECG's pricing system already reacts to demand: when a campsite fills up, prices rise, when bookings are slow, prices fall. In the historical record, high prices and high bookings therefore tend to occur together, not because price causes bookings, but because both respond to a common, unobservable underlying demand signal.

The fix requires identifying the demand signal that the pricer reacts to. Bookings on books, the cumulative number of nights already reserved at the moment a price is set, is observed in our data and lies on the only open backdoor path between price and remaining demand. Conditioning on it blocks the reverse-causation channel and better isolates the genuine effect of price.

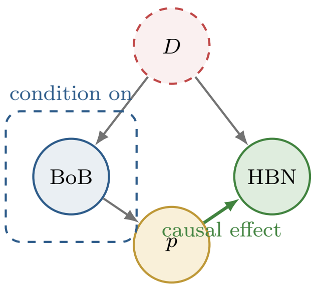

Figure 1: Causal structure linking price (p) and bookings (HBN). Latent demand D (red, dashed, unobservable) drives bookings-on-books (BoB, observed) and bookings directly. The existing pricer sets p in response to BoB. The path $p \leftarrow \text{BoB} \leftarrow D \rightarrow \text{HBN}$ is a backdoor: it lets demand contaminate the apparent price-bookings relationship. Conditioning on BoB (dashed blue box) closes it, leaving only the genuine causal effect $p \rightarrow \text{HBN}$.

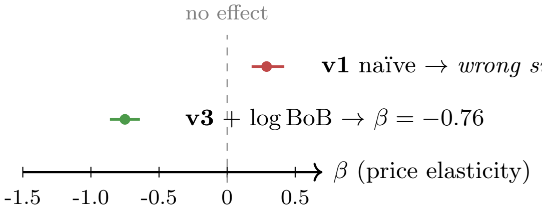

Figure 2: Posterior point estimate and 80% credible interval for the global price elasticity β. The naïve specification (v1) recovers an economically impossible positive sign, direct evidence of price endogeneity in ECG's data. After conditioning on bookings-on-books, the elasticity becomes negative as economic theory requires (v3, $\beta = -0.76$).

3 Two models, two jobs

The Bayesian estimate of $\beta = -0.76$ from model v1 implies demand is inelastic on average ($|\beta| < 1$). In plain language, every 1% price increase reduces bookings by about 0.76%. Because the response is less than proportional, total revenue (price multiplied by bookings), on average, goes up when prices go up, at least within the price range our data cover. Under a constant-elasticity assumption, this would mean ECG could raise prices uniformly and capture more revenue. A richer LightGBM optimiser in Section 4 refines this picture: average elasticity is right, but per-cell optima vary widely. Some campsite-week cells are over-priced, others under-priced, and the aggregate move is close to zero. The business case for a pricing engine on this dataset is therefore not "raise everything" but "identify the asymmetries", figuring out which cells should move, in which direction, and by how much.

Model v3 is intentionally minimal: it only uses the variables strictly needed to close the backdoor. This makes it a clean estimator of the local average elasticity but not a forecaster. To predict bookings at any candidate price for any single snapshot of the booking horizon, we train a gradient-boosted-trees model (LightGBM) on the same conditioning structure plus the lead-time, calendar and accommodation features that v3 deliberately omits. Trees can include all of these without the over-conditioning collapse that breaks richer Bayesian specifications. At typical prices, the LightGBM model implies the same local price-bookings slope as the Bayesian model (around -0.75), a useful cross-check: two methods with very different assumptions arrive at the same answer.

Out-of-sample, LightGBM achieves a median ROMGID-level error of 27% on total booked nights, within industry norms. The error is concentrated in low-volume ROMGID's where small absolute differences inflate percentage errors. For ROMGID's with at least 10 bookings per season, median error drops to 22%.

4 The optimiser and what it recommends

For each test-fold snapshot independently, we ask the LightGBM forecaster a counterfactual question: at this point in the booking horizon, what price would have produced the highest predicted revenue? We sweep 21 candidate price multipliers in the range $[0.85, 1.15] \times p_{\text{obs}}$, predict bookings at each candidate, compute the implied revenue $p \cdot \hat{N}(p)$, and pick the maximum. The output for a single ROMGID is therefore not a single static price but a trajectory of 53 prices across the booking horizon, one per snapshot.

Three constraints keep the recommendation defensible: the candidate range is capped at $\pm 15\%$ of historical prices for that group, predicted bookings are clipped at capacity, and every recommendation passes through a two-tier flag (green / yellow) before deployment, keeping the human in the loop.

Applied to 14,200 ROMGID's in the held-out test fold:

Metric	**Value**
Median per-ROMGID Δprice	-0.7%
Δprice IQR (per ROMGID)	$[-6.0\%, +5.4\%]$
% snapshots at upper bound (1.15×)	12.7%
% snapshots at lower bound (0.85×)	14.7%
% snapshots with interior optimum	**72.6%**
Total predicted uplift	€63M
Uplift as % of baseline	+38.8%
Flag split (green / yellow)	51% / 49%

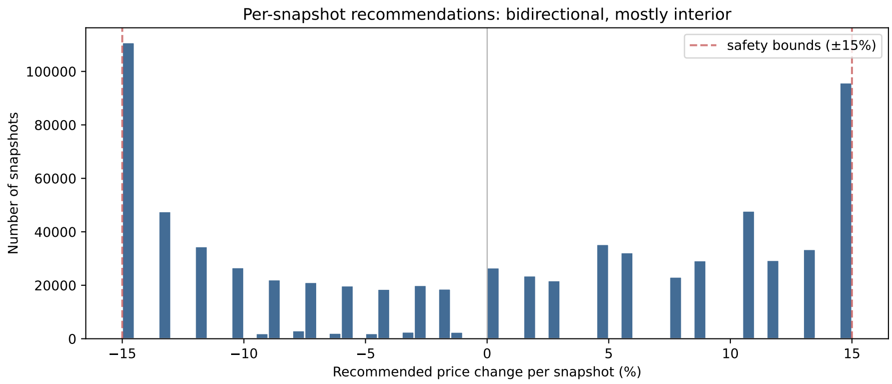

Figure 3: Distribution of recommended price changes across all test-fold snapshots. The histogram is roughly symmetric around zero, with mass at both the lower and upper safety bounds. About 73% of decisions land at an interior optimum within the safe range, 14.7% want a 15% discount, and 12.7% want a 15% increase.

The most important number on the page is the **72.6% interior-optimum rate**. For nearly three quarters of snapshot decisions, the model finds a revenue-maximising price strictly inside the ±15% safe range, not at a boundary. This is something the constant-elasticity Bayesian model in Section 2 could not produce. When elasticity is forced constant the revenue function has no interior maximum, but when LightGBM learns a flexible price-demand curve, genuine interior optima emerge. The recommendations are also bidirectional, 14.7% of snapshots want a price decrease and 12.7% want the maximum 15% increase. ECG is over-pricing some cells and under-pricing others, with the bulk of decisions sitting at modest interior adjustments.

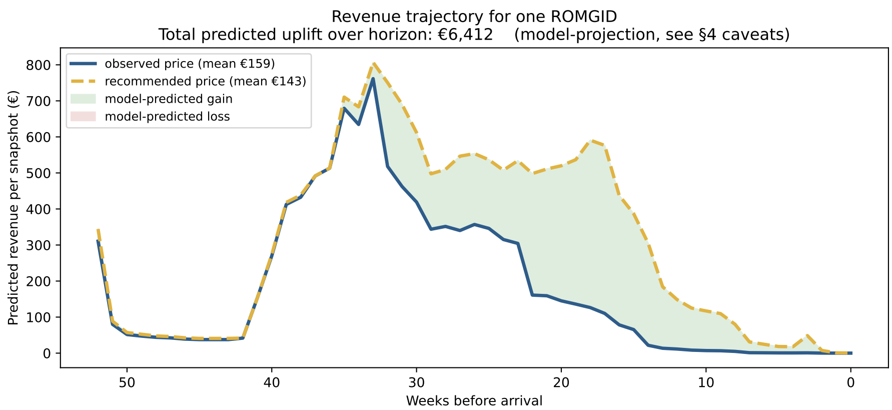

Figure 4: Predicted revenue per snapshot for one representative ROMGID, observed price (blue) versus recommended price (amber), with 80% prediction intervals from a quantile LightGBM. Most of the booking horizon shows similar predicted revenue under both strategies, with the recommended trajectory slightly higher in the mid-horizon weeks where bookings actually concentrate.

The +€63M total predicted uplift is a model projection and is almost certainly optimistic, for three reasons. First, the model has 27% out-of-sample error on observed prices, and counterfactual prices push it further from familiar ground. Second, the optimiser picks the candidate that maximises predicted revenue, exploiting any residual model error in its own favour. Third, the model can predict small positive bookings even in late-horizon snapshots where actual bookings are essentially zero, and the optimiser sees these as small revenue gains. The directional finding (interior optima, modest adjustments, bidirectional recommendations) is robust, but the magnitude is not. A randomised price experiment, discussed next, is the only way to convert it into a number ECG can budget against.

5 What ECG should do next

Two complementary tracks follow from this report. The first validates the magnitude. The second deploys the workflow.

1. Run a small randomised price experiment. The +€63M figure is a model projection, not a calibrated lift estimate. The three reasons it is optimistic, listed at the end of Section 4, cannot be resolved by further analysis of the historical data. They can only be resolved by generating new data under a known, randomised intervention. We recommend selecting 100 to 200 ROMGID's across markets and accommodation types, randomising their price by ±25% over a single 30-week season, and observing the resulting bookings. This delivers two things at once: an unbiased elasticity estimate to refit the model on, and an extension of the recommendation envelope beyond the ±15% safe-range cap that currently bounds the optimiser. The cost is bounded uncertainty on a small subset of inventory; the return is a deployable, calibrated pricing system.

2. Run the pipeline in shadow mode alongside the experiment. The pipeline produces a `recommendations.csv` every time it runs, with one row per ROMGID containing the current price, the recommended price, predicted bookings, revenue uplift with an 80% credible interval, and a human-in-the-loop flag. Because the model conditions on bookings-on-books at the moment of prediction, any new reservation that enters ECG's booking system automatically shifts the ROMGID to a different point on the demand curve. A nightly batch job recomputes features for all active ROMGID's, reruns the price sweep, and regenerates the file, so the dashboard always reflects the most recent booking state without requiring model retraining. Retraining itself happens once per season, when a full new round of observations is available to update the elasticity estimates.

The prototype dashboard we deliver alongside this report reads the recommendation file and presents each ROMGID as a row in a sortable, filterable table. A pricing manager opening the dashboard sees, for every campsite-week combination, the current price next to the model's recommended price, the percentage change, the predicted revenue uplift, and a colour-coded flag. Clicking a row expands a detail panel with observed versus predicted bookings, a capacity-utilisation bar, historical price volatility, and a plain-language explanation of why this recommendation was flagged the way it was.

ROMGID	**Campsite**	**Type**	**Flag**	**Current**	**Recommended**	**Δprice**
0042	CS-FR-118	Mobile Home	yellow	€187	€164	−12.3%

Figure 5: Example row from the pricing dashboard. This ROMGID is flagged yellow because the recommended adjustment exceeds 10%; the pricing manager reviews local context before approving.

The flag system connects the model to the human. A **green** flag means the recommended adjustment is small (at most 10%) and the historical price has been stable (coefficient of variation below 0.15). A **yellow** flag means the adjustment is larger (10–20%) or the historical price has been moderately volatile; the model's recommendation is plausible but needs a manager to check what the model cannot see, a local festival, an incoming weather front, a competitor's flash sale. Yellow recommendations form the manager's daily review queue.

Why both tracks are necessary. The two tracks answer different questions. The experiment validates the *magnitude* of the recommended adjustments by generating randomised counterfactual data the historical record cannot provide. Shadow mode builds the *managerial trust* required for adoption, by letting the team compare the engine's suggestions against their own decisions over a season and develop a feel for where the model adds value and where domain expertise remains essential. Selective auto-application of green-flag recommendations should follow only once both signals are positive: the experiment confirms the elasticity at the ROMGID level, and shadow-mode track records show the model is making sensible recommendations in segments managers trust. Until the experimental data are in, every recommendation remains advisory. The manager's role then shifts from setting every price to auditing the exceptions, a more efficient use of their domain knowledge.

Limitations. The methodology rests on three assumptions worth flagging. First, we close one back-door (demand-autocorrelation via $\log \text{BoB}$). Other sources of price endogeneity, such as channel-mix changes and manager overrides, remain unobservable in this dataset. Second, LightGBM's 27% out-of-sample error grows for counterfactual prices the model has not seen, and the optimiser exploits that error in its own favour. Third, capacity is structurally non-binding here (median utilisation 10%); a real deployment in peak weeks at premium properties would encounter the capacity constraint that is currently a tested but inactive safeguard.

6 Conclusion

Treating dynamic pricing as a causal-inference problem reveals a clear endogeneity trap and a defensible local elasticity of -0.76 once it is closed. A complementary LightGBM forecaster delivers per-snapshot recommendations that are differentiated and bidirectional, identifying which cells should move and in which direction. The outcome is not just a model but a working pipeline: a recommendation file, a flag-based review workflow embedded in a dashboard, and a phased integration plan in which a one-season randomised experiment validates the lift magnitude while shadow-mode operation builds managerial trust. The engine does not replace the pricing manager; it focuses their attention on the ROMGID's where intervention matters most, while preserving their existing expertise rather than overruling their judgement entirely. This aspect is also important in convincing the existing managers to adopt the project rather than perceiving it as a threat to their role.

ECG's status quo is likely leaving model-identifiable revenue on the table. The size of that revenue is the open question this report cannot answer alone, but a small randomised experiment, run alongside shadow-mode deployment of the pipeline, is designed to answer it within one season at bounded cost.